

24**B**

Force on electron = $Bqv \sin \theta$

$$= (2.0 \times 10^{-3})(1.60 \times 10^{-19})(3.0 \times 10^6) \sin 90^\circ = 9.6 \times 10^{-16} \text{ N}$$

Electron undergoes uniform circular motion in magnetic field as magnetic force provides centripetal force. So magnitude of velocity does not change, and hence kinetic energy does not change. Gain in kinetic energy is zero.

Velocity of electron changes as direction changes. Hence, change in momentum is not zero.